LAB Logbook

Lab 1

Data Frame – A 2-dimensional labelled data structure that holds data in rows and columns, similar to a table in Excel or SQL.

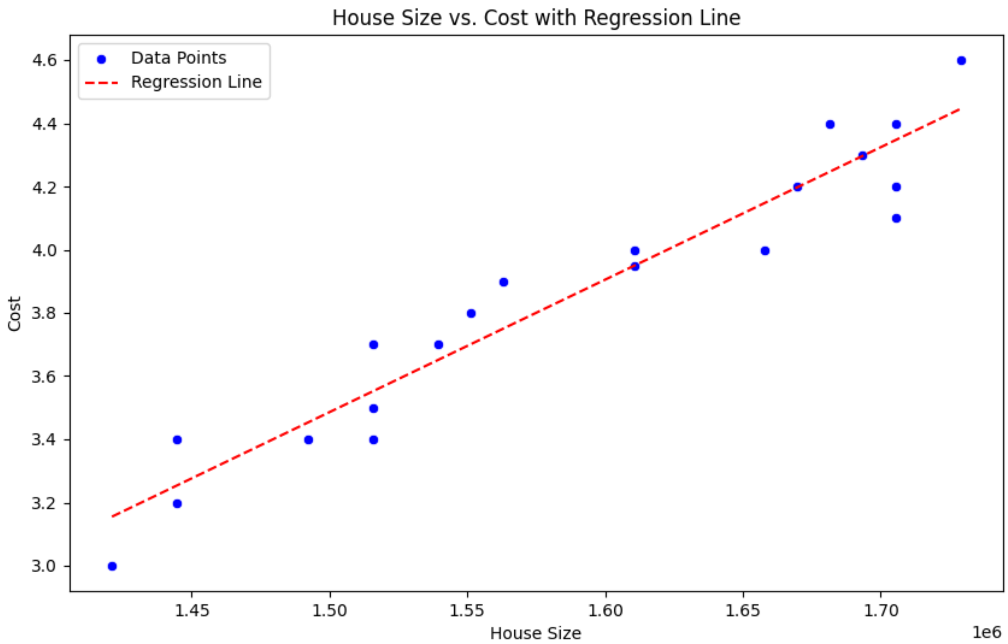
Series – A one-dimensional array-like object that can hold data of any type and includes axis labels.

Index – An immutable object that labels the axes of a Data Frame or Series and supports quick lookups and alignment.

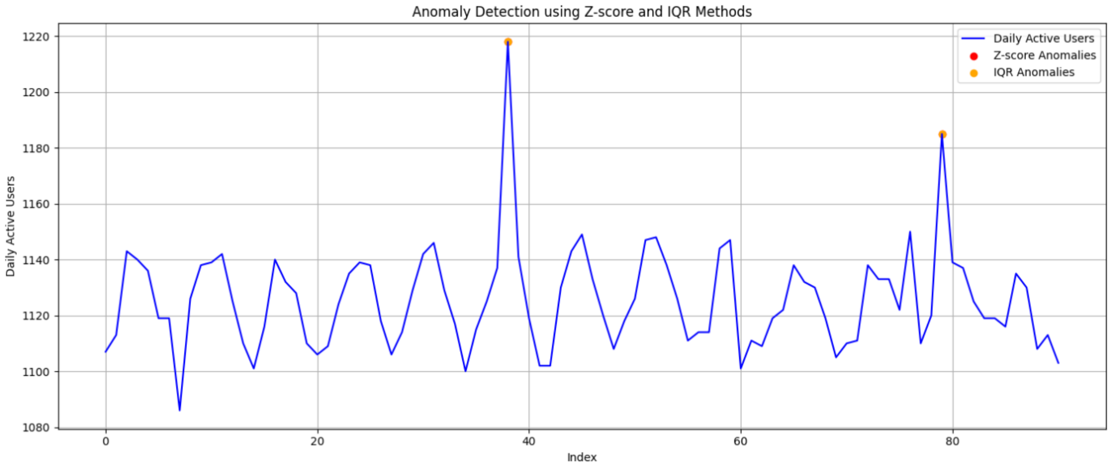
Categorical – A class for working with variables that take on a limited, fixed number of possible values, optimizing memory and speed.

Timestamp – A powerful class for representing single points in time, built on top of Python's datetime, used in time series data.

Lab 2

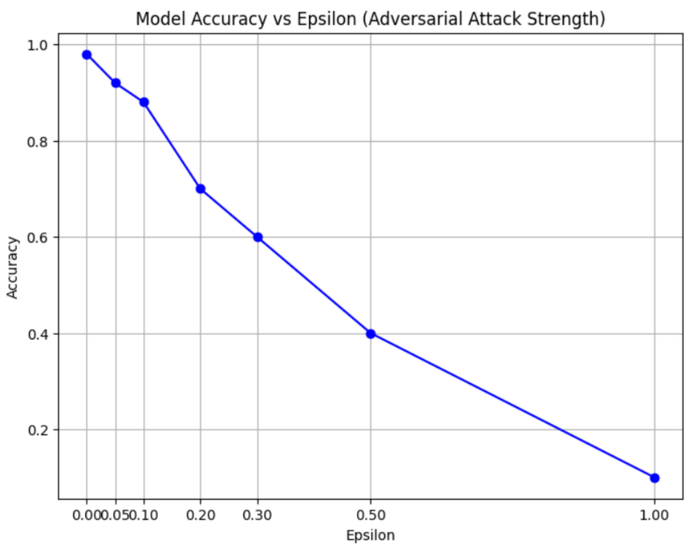


Lab 3

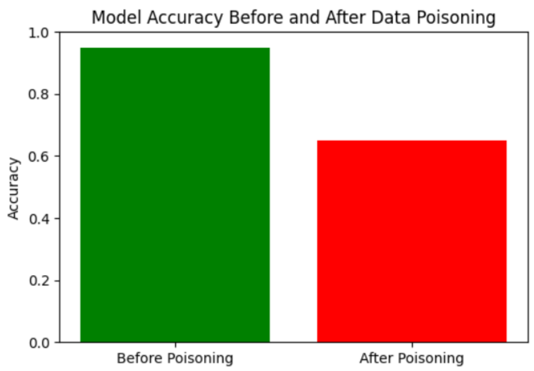


Anomaly Detection using Z-score and IQR Methods
Daily Active Users
Z-score Anomalies
IQR Anomalies
Daily Active Users
1220
1200
1180
1160
1140
1120
1100
1080
0
20
40
60
80
Index

Lab 4

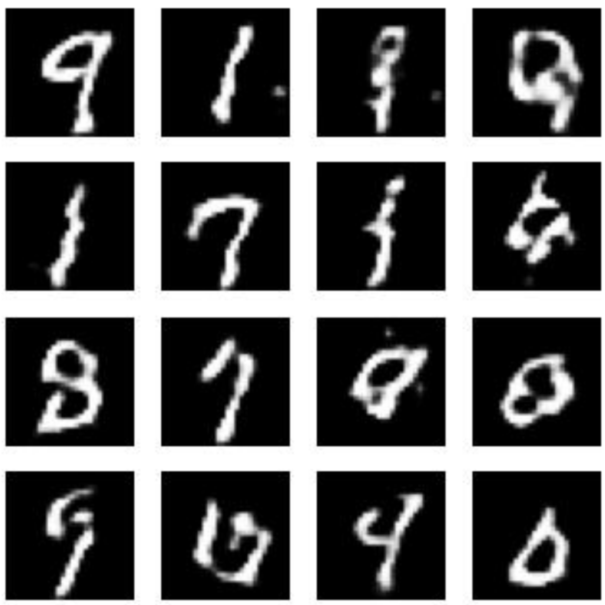


Model Accuracy vs Epsilon (Adversarial Attack Strength)
Accuracy
1.0
0.8
0.6
0.4
0.2
0.00
0.05
0.10
0.20
0.30
0.50
1.00
Epsilon



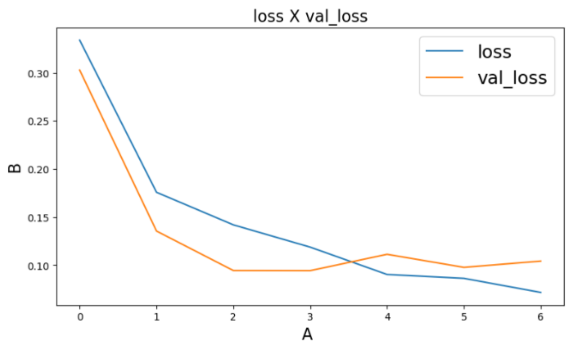
Lab 5

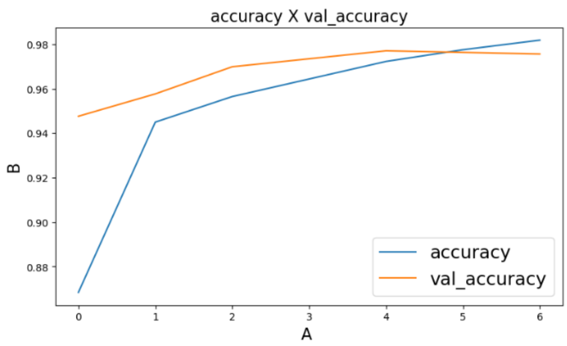
```
Time for epoch 34 is 69.01496171951294 sec
```



```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 407ms/step
Screenshot saved at epoch 29, training time: 0.00 sec
```

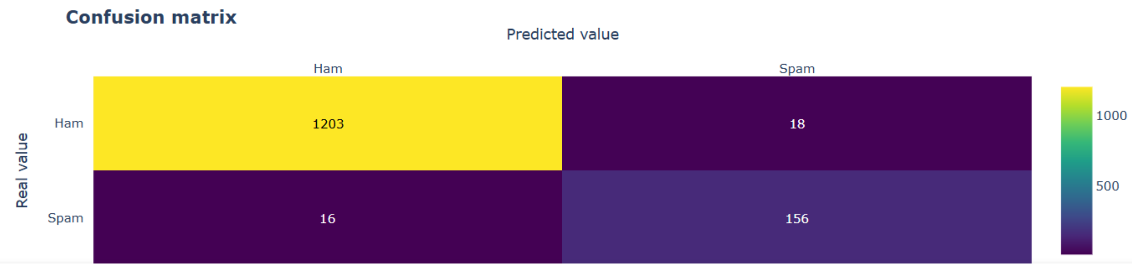
Lab 6






```
44/44 ━━━━━━━━━━━━━━━━━━━━ 5s 95ms/step
```

Collapse Output

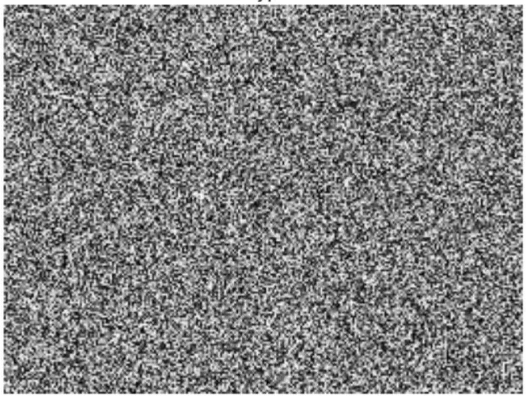


Lab 7

```
Encrypted image saved as encrypted_ecb.png
```

Original

Encrypted



3. Explain in one word - 'YES' or 'NO' whether your encryption method for the images is good.

NO

The encryption method is not secure because visible patterns from the original image are still present in the encrypted version.

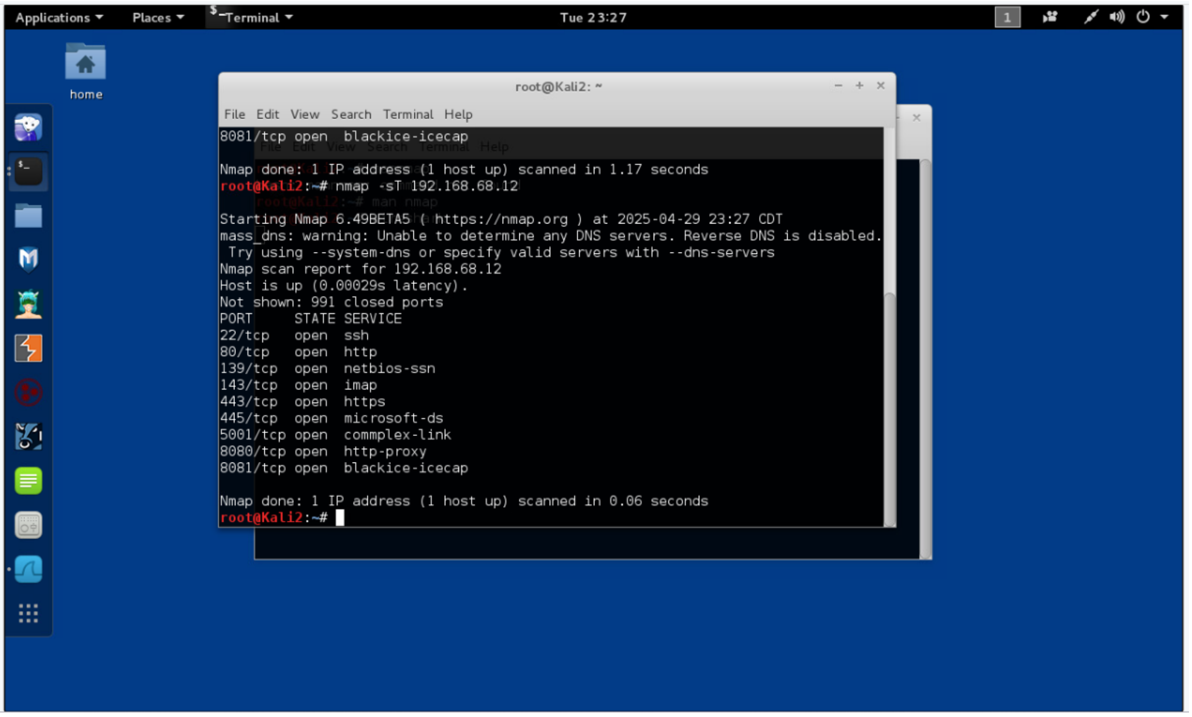
Lab 8

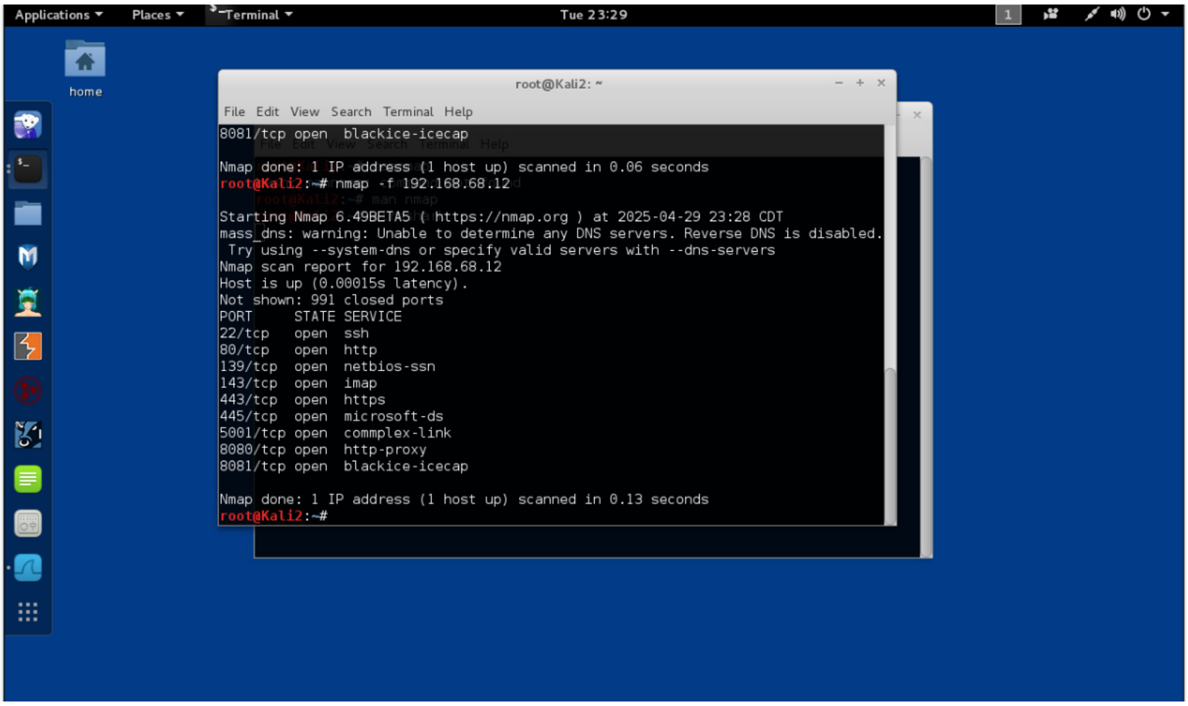
1. Partner's Name: Azad
2. Public Values:
3. p = 23
4. g = 5
5. My Private Key: a = 6

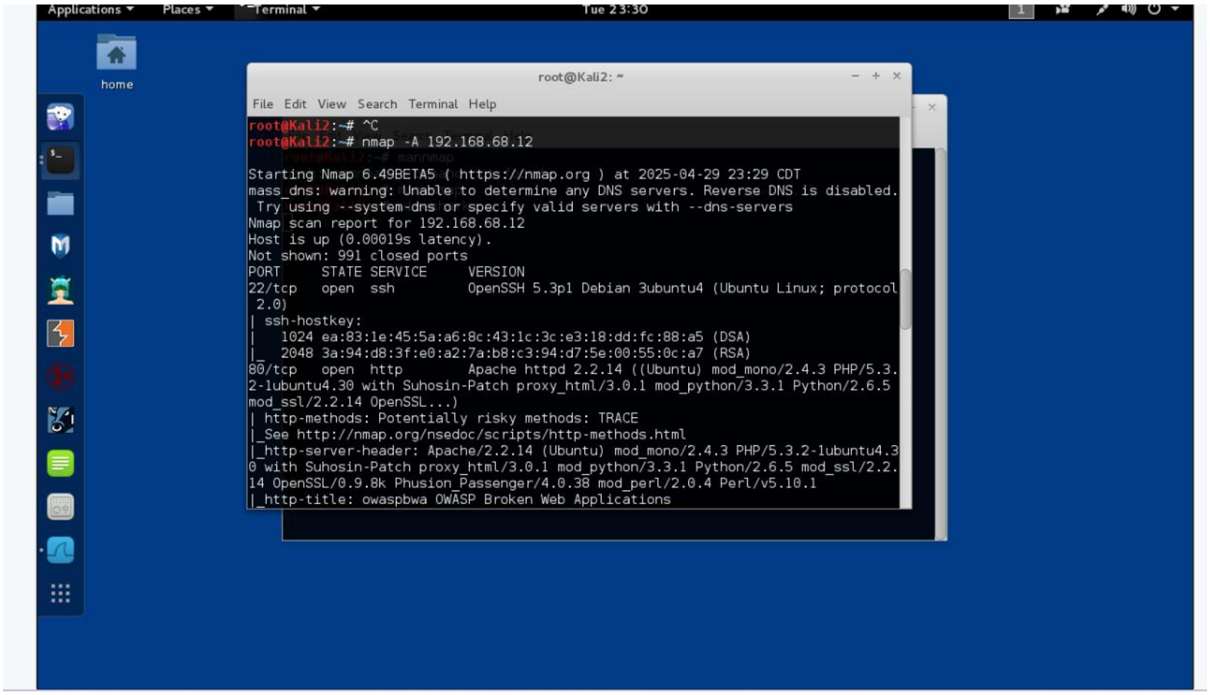
Lab 9

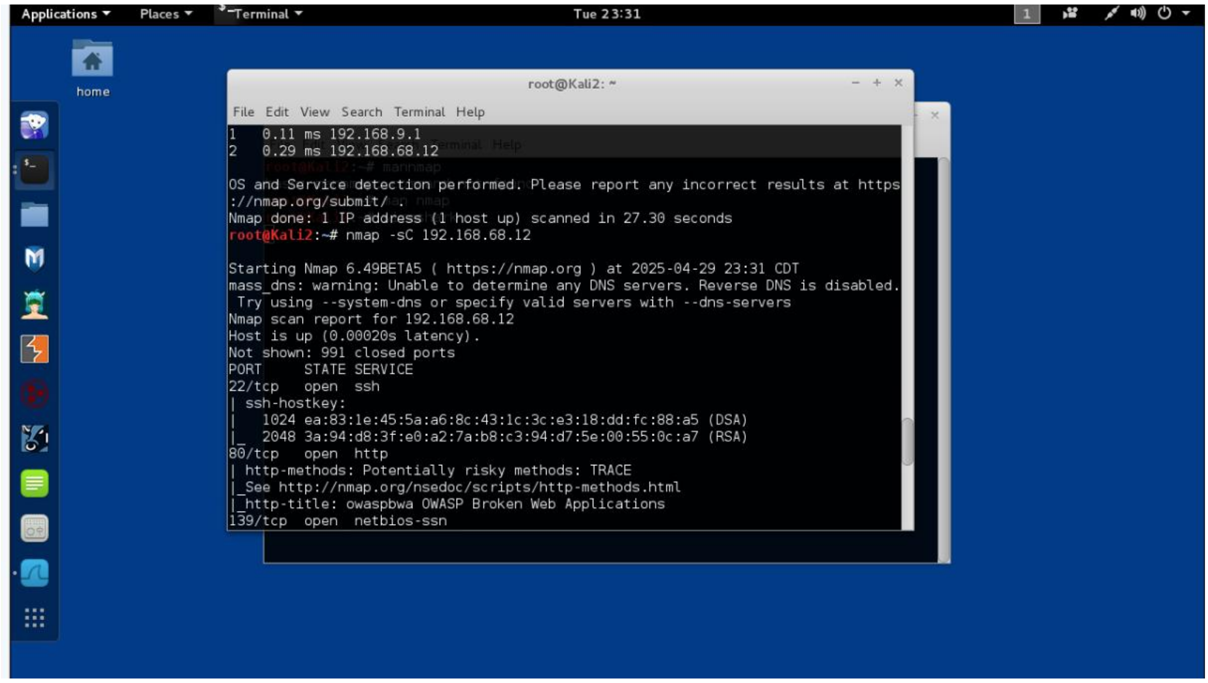
Attack Type Chosen: Log4j Vulnerability Exploit (2021)
Key Website/Research Link:
https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-356a

Lab 10


Applications Places Terminal Tue 23:27
home
root@Kali2: ~
File Edit View Search Terminal Help
8081/tcp open blackice-icecap
Nmap done: 1 IP address (1 host up) scanned in 1.17 seconds
root@Kali2:~# nmap -sT 192.168.68.12
Starting Nmap 6.49BETA5 (https://nmap.org) at 2025-04-29 23:27 CDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.68.12
Host is up (0.00029s latency).
Not shown: 991 closed ports
PORT   STATE SERVICE
22/tcp   open ssh
80/tcp   open http
139/tcp open netbios-ssn
143/tcp open imap
443/tcp open https
445/tcp open microsoft-ds
5001/tcp open commplex-link
8080/tcp open http-proxy
8081/tcp open blackice-icecap
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
root@Kali2:~#


Applications Places Terminal Tue 23:29
home
root@Kali2: ~
File Edit View Search Terminal Help
8081/tcp open blackice-icecap
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
root@Kali2:~# nmap -f 192.168.68.12
Starting Nmap 6.49BETA5 (https://nmap.org) at 2025-04-29 23:28 CDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.68.12
Host is up (0.00015s latency).
Not shown: 991 closed ports
PORT   STATE SERVICE
22/tcp   open ssh
80/tcp   open http
139/tcp open netbios-ssn
143/tcp open imap
443/tcp open https
445/tcp open microsoft-ds
5001/tcp open commplex-link
8080/tcp open http-proxy
8081/tcp open blackice-icecap
Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
root@Kali2:~#

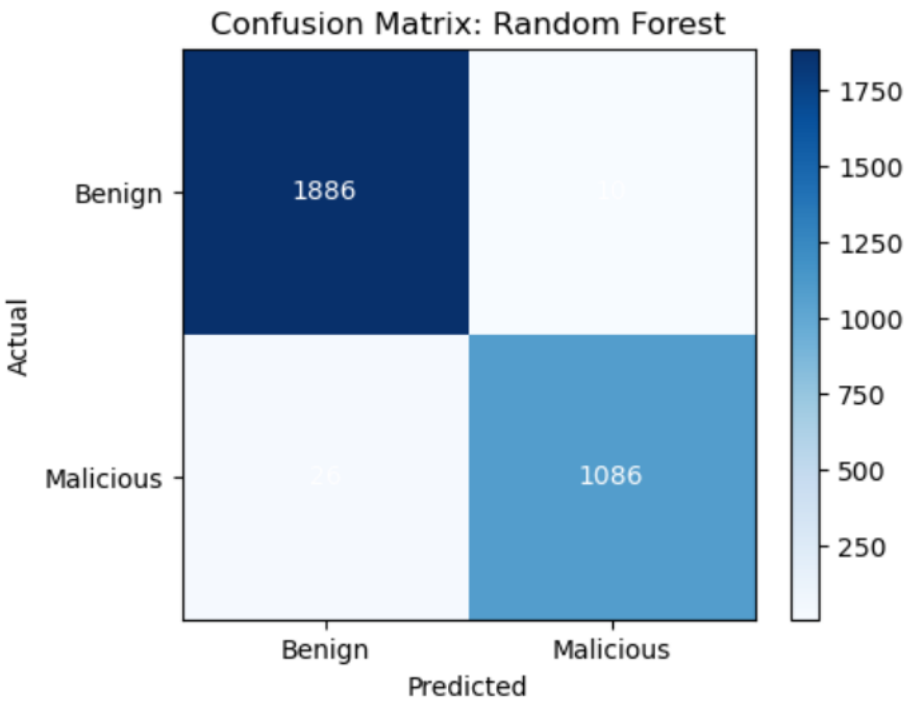

Applications
Places
Terminal
Tue 23:30
home
root@Kali2: ~
File Edit View Search Terminal Help
root@Kali2:~# ^C
root@Kali2:~# nmap -A 192.168.68.12
Starting Nmap 6.49BETA5 (https://nmap.org) at 2025-04-29 23:29 CDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.68.12
Host is up (0.00019s latency).
Not shown: 991 closed ports
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 5.3p1 Debian 3ubuntu4 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 1024 ea:83:1e:45:5a:a6:8c:43:1c:3c:e3:18:dd:fc:88:a5 (DSA)
|_ 2048 3a:94:d8:3f:e0:a2:7a:b8:c3:94:d7:5e:00:55:0c:a7 (RSA)
80/tcp open http Apache httpd 2.2.14 ((Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL...)
| http-methods: Potentially risky methods: TRACE
|_See http://nmap.org/nsedoc/scripts/http-methods.html
|_http-server-header: Apache/2.2.14 (Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL/0.9.8k Phusion_Passenger/4.0.38 mod_perl/2.0.4 Perl/v5.10.1
|_http-title: owaspbwa OWASP Broken Web Applications


Applications
Places
Terminal
Tue 23:31
home
root@Kali2: ~
File Edit View Search Terminal Help
1 0.11 ms 192.168.9.1
2 0.29 ms 192.168.68.12
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 27.30 seconds
root@Kali2:~# nmap -sC 192.168.68.12
Starting Nmap 6.49BETA5 (https://nmap.org) at 2025-04-29 23:31 CDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.68.12
Host is up (0.00020s latency).
Not shown: 991 closed ports
PORT STATE SERVICE
22/tcp open ssh
| ssh-hostkey:
| 1024 ea:83:1e:45:5a:a6:8c:43:1c:3c:e3:18:dd:fc:88:a5 (DSA)
|_ 2048 3a:94:d8:3f:e0:a2:7a:b8:c3:94:d7:5e:00:55:0c:a7 (RSA)
80/tcp open http
| http-methods: Potentially risky methods: TRACE
|_See http://nmap.org/nsedoc/scripts/http-methods.html
|_http-title: owaspbwa OWASP Broken Web Applications
139/tcp open netbios-ssn

Lab 11

```
=== Model Performance Comparison ===
```

	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.978391	0.973756	0.967626	0.970681
1	Random Forest	0.988032	0.990876	0.976619	0.983696
2	SVM (RBF Kernel)	0.982048	0.980909	0.970324	0.975588



<u>Lab 12</u>